

Power law catchment-scale recessions arising from heterogeneous linear small-scale dynamics

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Received 27 August 2008; revised 2 May 2009; accepted 20 May 2009; published 9 September 2009.

[1] Power law recession behavior has typically been attributed to the hydraulics of subsurface flows in the landscape (e.g., the Boussinesq equation). Many catchments exhibit enormous heterogeneity in their characteristics, such as the permeability of soils, hillslope length and gradient, and soil mantle thickness. In the paper we demonstrate using a simple linear flow model that the between-hillslope heterogeneity alone can give rise to observed apparently nonlinear recession slope curves. The degree of nonlinearity (represented by the exponent fitted to the recession slope curve) is shown to increase with the degree of heterogeneity of catchment hydraulic properties. In sufficiently heterogeneous catchments, the recession is also sensitive to the history of recharge events, tending to increase when the past events have built up a storage of water in the ‘slow’ parts of the catchment. The models based on the catchment heterogeneity were fitted to data from the Panola Mountain Research watershed, yielding a good fit, and allowing us to infer how the heterogeneity in catchment properties varies with scale. The range of exponents explained by this theory overlaps with the range explained by the hydraulics of subsurface flow, which raises doubts about our ability to extract catchment hydraulic parameters on the basis of recession curves so long as the heterogeneity is not accounted for. In general the results presented here have enabled us to show how the heterogeneity of landscape characteristics is upscaled and manifested in the apparent nonlinear relationship at the catchment scale.

Citation: Harman, C. J., M. Sivapalan, and P. Kumar (2009), Power law catchment-scale recessions arising from heterogeneous linear small-scale dynamics, *Water Resour. Res.*, 45, W09404, doi:10.1029/2008WR007392.

1. Introduction

1.1. Recession Curves and Catchment Properties

[2] The patterns, or ‘signatures’, present in hydrographs offer vital, if narrow, windows into the hydrologic functioning of catchments, but their interpretation depends on the conceptual model adopted [Sivapalan, 2005]. One such signature is the recession curve, which refers to the shape of the decline of measured discharge some time after the storm has ended. It has been observed that the rate of change of discharge dQ/dt during the recession at a given discharge Q is often consistent or invariant between storm events [Brutsaert and Nieber, 1977]. This suggests that it reflects fundamental hydrologic features of the catchment in some way. The relationship $-dQ/dt = g(Q)$ has been called the ‘recession slope curve’ [Rupp and Selker, 2006b]. It has often been observed that the recession slope curve follows a power law of the form $g(Q) = aQ^b$. The exponent b and coefficient a vary from catchment to catchment [Brutsaert and Nieber, 1977]. Observations of b range from less than 1

to larger than 3 [Kirchner, 2009; Tague and Grant, 2004; Chapman, 1999; Wittenberg, 1999]. Perhaps because it is simple, easy to estimate and ubiquitous, recession slope curves have had a significant influence on our understanding of catchment processes like subsurface discharge to rivers and on the models used to encode that understanding and make predictions [Brutsaert and Lopez, 1998; Szilagyi et al., 1998; Shaw et al., 2008; Rupp and Selker, 2006a; Mendoza et al., 2003; Sloan, 2000; Manga, 1999; Moore, 1997; Tallaksen, 1995; Troch et al., 1993a; Vogel and Kroll, 1992; Zecharias and Brutsaert, 1988].

[3] The physical origins of the recession slope curve have been subject to considerable study in order to determine whether information about catchment properties can be extracted from them. A common explanation is that they arise from the hydraulics of flow during the discharge of water from a hillslope or riparian aquifer [Brutsaert and Nieber, 1977; Brutsaert, 1994; Rupp and Selker, 2006b; Brutsaert, 1994]. This explanation is based on derivations from the Boussinesq equation, which can describe saturated flow over a sloping base, and assumes that the catchment-scale behavior follows that of an ‘effective hillslope’ whose properties are representative (in some sense) of the properties of the collection of landforms that make up a catchment. It is well known, however, that catchment properties are highly heterogeneous, both within and between hillslopes [Troch et al., 2008]. Subsurface discharge from a hillslope depends on the hydraulic and topographic gradients driving flow, soil properties, topography, temporal variability of the

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recharge from the unsaturated zones, the boundary condition at the base of slope and other factors [Henderson and Wooding, 1964; Beven, 1982; Brutsaert, 1994; Harman and Sivapalan, 2009a]. A number of studies have investigated the role of internal heterogeneity in these properties on the subsurface discharge from hillslopes, and found that it is indeed significant [Rupp and Selker, 2005; Fiori and Russo, 2007; Harman and Sivapalan, 2009b]. However there have been fewer studies of the effects of variability between hillslopes on the shape of the recession slope curve at the catchment scale.

[4] In a recent study of recessions for three nested catchments in the Panola Mountain Research Watershed, Clark *et al.* [2009] showed that variability between hillslope properties was not only significant, it was likely to explain the shape of recession slope curve, and also why this shape changed with scale. They found that recession slope curves from an individual hillslope (0.1 ha) had an exponent $b = 1$, suggesting that it in fact behaved as a linear reservoir. However at larger scales (10 ha and 41 ha) the recession slope curves became increasingly nonlinear. Clark *et al.* [2009] showed that it was possible to reproduce this behavior, while maintaining some consistency in the process descriptions, using a model in which discharge from two or three linear reservoirs is summed. Each linear reservoir represented a different component of the landscape (hillslopes, ephemeral riparian aquifers, permanent stream aquifers) and each had a different characteristic timescale, and a different fractional area. Clark *et al.* [2009] found that higher observed exponent could be fitted using a broader range of timescales.

[5] The observations of Clark *et al.* [2009] suggest that the observed b exponents could be attributed at least partly to the heterogeneity of catchment properties. However, the link between the recession exponent and the set of characteristic timescales that represent this variability is not quantified explicitly.

[6] Also, in Clark *et al.* [2009] the empirical recession data from the catchment was fitted to modeled data representing the recession following steady state recharge. Field estimates of the b exponent are derived from recession data following individual storms, not steady state recharge [Clark *et al.*, 2009; Zecharias and Brutsaert, 1988; Chapman, 2003]. They therefore reflect the response to recharge of a variety of durations and intensities, and a variety of initial conditions. The influence of this temporal variability on b has not been examined in the context of catchment spatial heterogeneity.

1.2. Aims of This Work

[7] In this work we aim to develop a general theory based on the conceptual model of Clark *et al.* [2009] that provides a quantitative link between the variability of catchment properties and the recession slope curve exponent b . We will also examine the role of the temporal variability of recharge in controlling this exponent.

[8] We will do so by assuming that the heterogeneity of catchment properties can be captured by a probability density function (pdf) of the characteristic timescales. This simplified conceptual model allows us to develop a simple mathematical theory of discharge from heterogeneous catchments for arbitrary probability distributions. This theory can then be used to establish the connection between the

shape of the recession slope curve, the properties of the probability distribution and the type of recharge variability. In particular the link between the exponent b and the coefficient of variation ($CV = \text{mean}/\text{std deviation}$) related to the between-hillslope heterogeneity will be examined. The approach we adopt is similar to methods in statistical mechanics known as “superstatistics” for describing the energy states of driven, spatially variable systems [Touchette and Beck, 2005; Beck and Cohen, 2003]. This method essentially involves the randomization of a parameter in a probability distribution relevant to the process according to another distribution that accounts for variability in the catchment properties controlling the process. This method has been used extensively in other disciplines, such as chemical engineering, and even ecology [Thompson and Katul, 2008]. However, it has not been used in this particular context previously.

[9] Since we do not know what the appropriate distribution of properties is for typical catchments, common and mathematically convenient distributions will be discussed, and methods used in the literature on superstatistics will be used to obtain results for a general class of power law tailed distributions. We investigate the effect of temporal variability of recharge on the relationship between b and the catchment heterogeneity first by examining two end-member cases for which analytical expressions can be derived (the recession following a single impulse, and the recession following steady state recharge) and second examining the response to periodic recharge events.

[10] The theory is then tested against the recession slope curves at three catchment scales presented by Clark *et al.* [2009] from the Panola watersheds. We use this theory to examine how the variation in catchment properties varies with scale by fitting the derived expressions for the recession slope curve to the observations, under different assumptions of the mode of recharge forcing and type of distribution of timescales. We also compare the ability of catchment heterogeneity and hydraulic theory to explain the ranges of b exponents reported in literature from catchments around the world. Finally we discuss the implications for catchment modeling and analysis.

2. Theory

2.1. Linear Reservoir Approximation

[11] Let us assume that the catchment recession discharge can be approximated by a collection of individual hillslopes that behave like linear reservoirs with various characteristic timescales, arranged in parallel. In a linear reservoir the discharge Q is a linear function of storage S as $Q = S/T$ or $Q = VS$. The timescale T , or alternatively $1/V$, is the characteristic timescale of the response. The observation of Clark *et al.* [2009] provides some empirical justification for this approach. It has also been shown that under certain circumstances the hydraulic theory of subsurface discharge from hillslopes and riparian aquifers does follow this relationship, and in such cases the timescale is proportional to the hillslope length L and drainable porosity ϕ_d , and inversely proportional to the slope Θ and hydraulic conductivity K [Harman and Sivapalan, 2009a; Berne *et al.*, 2005]:

$$\frac{1}{V} \sim \frac{L\phi_d}{K \sin \Theta} \quad (1)$$

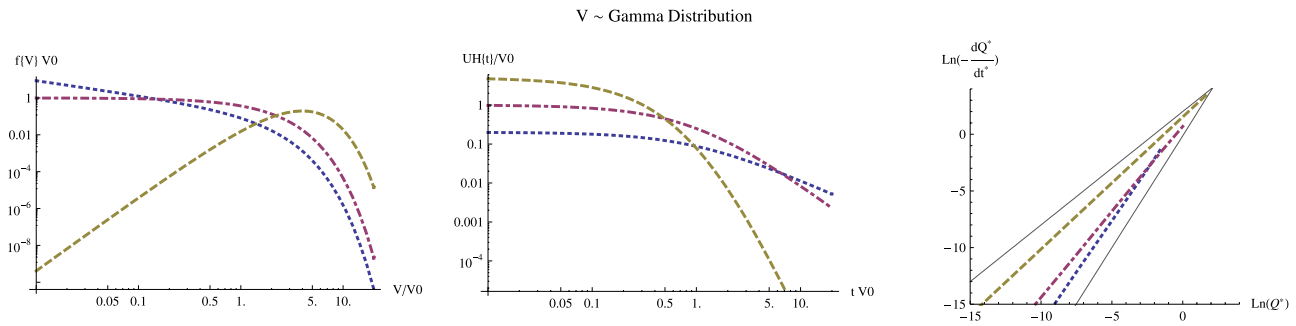


Figure 1. The (left) pdf $f(V)$, (middle) unit hydrograph $IUH(t)$, and (right) recession slope curve $g(t)$ for the case where $f(V)$ is a Gamma distribution (equation (5)). All axes are made dimensionless using the V_0 parameter. Values of α are 0.2 (dotted line), 1 (dash-dot), and 5 (dashed line). Example power laws with exponents 1 and 2 are plotted in gray in Figure 1 (right).

The variation in these properties within the catchment can be captured by treating V as a random variable with a probability density function (pdf) $f(V)$ with support $[0, \infty]$,

[12] The impulse-response or ‘instantaneous unit hydrograph’ (IUH) of single linear reservoir (per unit contributing area) takes the form of an exponential distribution with a timescale $1/V$:

$$IUH_{LR}(t, V) = Ve^{-Vt} \quad (2)$$

The IUH function of the ensemble, $IUH(t)$ is the compound distribution of $IUH_{LR}(t, V)$ and $f(V)$ [Benjamin and Cornell, 1970]. For a finite number of reservoirs n of equal contributing area this is:

$$IUH(t) = \sum_{i=1}^n \frac{V_i}{n} e^{-V_i t} \quad (3)$$

where the values of V_i are distributed as $f(V)$. In the limit of large n this sum converges to the integral [Benjamin and Cornell, 1970]:

$$IUH(t) = \int_0^\infty f(V) V e^{-Vt} dV \quad (4)$$

[13] The formulation in equation (4) is similar to the ‘superstatistical’ description of a thermally heterogeneous (i.e., nonequilibrium) system used in statistical mechanics [Beck and Cohen, 2003; Touchette and Beck, 2005]. There the integrand is $f(\beta) e^{-\beta E}$, where β is the inverse temperature of a part of the system, and E is an energy state of a particle. The result of the integration is an effective distribution of energy states of the composed system $B(E)$, whose components have mean inverse temperatures distributed as $f(\beta)$. Here the pdf $f(V)$ represents the distribution of inverse timescales, and arises due to the heterogeneity of catchment properties.

[14] Discharge from the catchment can be found by convolving the IUH with the time series of recharge. If the catchment has fully drained since previous events and is subjected to a short periods of recharge, the recession will be similar to the shape of the IUH, scaled by the total recharge volume. However this will not be true if the catchment is subjected to extended recharge events, or does not drain fully between them.

2.2. Impulse Responses

2.2.1. Gamma and Lognormal Distributions

[15] A closed form solution for the IUH can be found when $f(V)$ is a Gamma distribution, which takes the form:

$$f(V) = \frac{1}{V_0 \Gamma(\alpha)} e^{-\frac{V}{V_0}} \left(\frac{V}{V_0}\right)^{\alpha-1} \quad (5)$$

where $V_0 > 0$ and $\alpha > 0$. Figure 1 (left) shows this distribution for several values of the shape parameter α . The solution to equation (4) in this case is:

$$IUH(t) = \alpha V_0 (1 + tV_0)^{-1-\alpha} \quad (6)$$

Figure 1 (middle) shows how the shape of these unit hydrographs varies with α . They take a finite value at $t = 0$ of αV_0 and then converge on a power law with exponent $-1 - \alpha$.

[16] The recession slope curve following an impulse of recharge can be found by convolving the IUH (equation (6)) with an impulse with total volume Q_0 , and eliminating the time variable between the result and its derivative. This yields:

$$-\frac{dQ}{dt} = Q_0 V_0^2 \alpha (\alpha + 1) \left(\frac{Q(t)}{Q_0 V_0 \alpha}\right)^{1+\frac{1}{\alpha+1}} \quad (7)$$

This is indeed a power law with a slope that varies between $b = 2$ for $\alpha \rightarrow 0$ and $b = 1$ for $\alpha \rightarrow \infty$. Figure 1 (right) shows the recession slope curves for several values of α . For a Gamma distribution, the coefficient of variation is given by $CV = \alpha^{-1/2}$, and so we can write an expression for the exponent b in terms of the CV as:

$$b = 1 + \frac{CV^2}{1 + CV^2} \quad (8)$$

This simple result links the observed recession slope curve directly to the variability of the catchment characteristic timescales. It shows that in this case b is 1 for $CV = 0$ and approaches 2 as the CV increases. The variation in b with CV is shown in Figure 2.

[17] Similar behavior can be observed for the lognormal distribution, which has often been used to de-

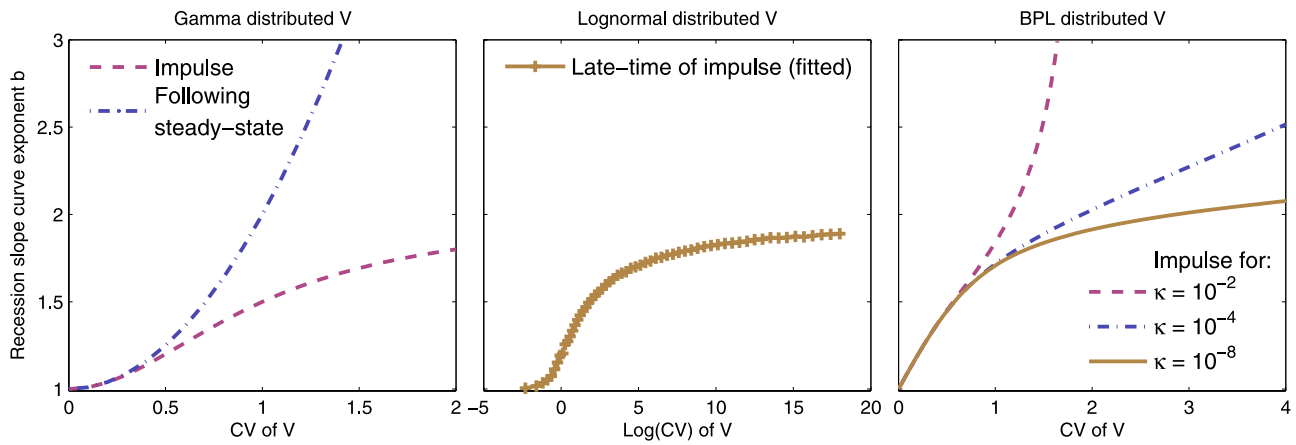


Figure 2. Derived relationships between variability in catchment properties and the recession slope curve exponent b . Curves are shown for the case of (left) a Gamma distribution following impulses (dashed line, equation (8)) and following steady state recharge (dash-dot line, equation (8)), for (middle) a lognormal distribution following an impulse (exponents estimated), and for (right) a bounded power law following an impulse (equations (21) and (25)) with various values of κ .

scribe the distribution of hydraulic conductivity values in catchments:

$$f(V) = \frac{1}{\sqrt{2\pi}V\sigma} e^{-\frac{(\log(V/V_0))^2}{2\sigma^2}} \quad (9)$$

Examples of this distribution for various values of σ are shown in Figure 3. However, a closed-form solution to the integral in equation (4) cannot be found with this distribution. We can plot the impulse response and recession slope curve though using equation (3) for large n with V_i drawn from the lognormal distribution, and examples are given in Figure 3. When σ is small the behavior is similar to a single linear reservoir, and the recession slope curve approaches $b = 1$. For larger σ the slope approaches $b = 2$, but does not exceed $b = 2$ except for a very brief period at the start of the recession. By fitting a power law to the late-time behavior for a range of values of the σ parameter, we can estimate the relationship between the CV and the recession slope curve exponent. The CV of a lognormal distribution is related to σ as:

$$CV = \sqrt{e^{\sigma^2} - 1} \quad (10)$$

The relationship between b and the CV is shown in Figure 2. The value of b at each CV was found to be insensitive to the value of Q_0 and V_0 .

2.2.2. General Impulse-Response Behavior for $f(V)$ With Power Law Tails

[18] We can extend these observations to a general class of distributions of V using methods suggested by the work on superstatistics by *Touchette and Beck* [2005]. That work derived expressions for the general classes of asymptotic behavior of ‘effective’ Boltzmann factors, but because of the extra factor of V in equation (4), their results are not directly applicable here, though the methods still are.

[19] Because the fast ($V \rightarrow \infty$) parts of the catchment will drain quickly, the discharge at late time ($t \rightarrow \infty$) will depend on the behavior of the ‘slow’ tail of the distribution ($V \rightarrow 0$). We can determine this behavior by noting that equation (4) takes the form of a Laplace integral, and using the Laplace, or ‘saddle-point’ approximation to estimate the value of the integral for large t [*Bender and Orszag*, 1978]. We begin by rewriting equation (4) as:

$$IUH(t) = \int_0^\infty e^{\Phi(V,t)} dV \quad (11)$$

where

$$\Phi(V,t) = -Vt + \log[Vf(V)] \quad (12)$$

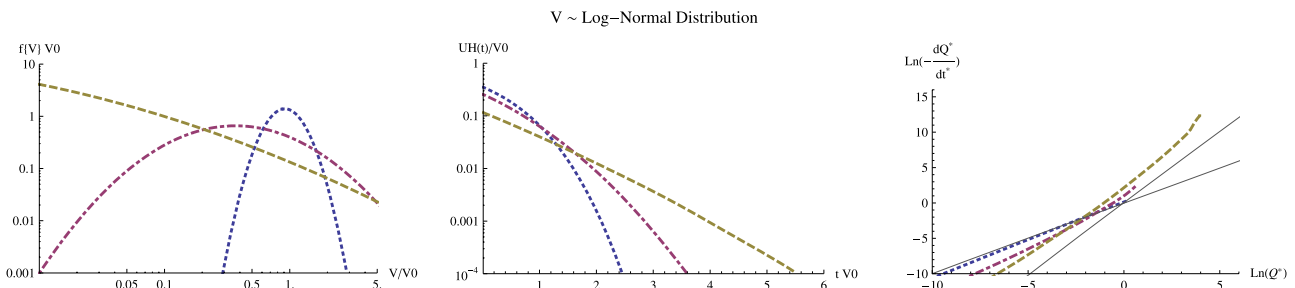


Figure 3. Same as Figure 1, but for the lognormal distribution (equation (9)). Values of σ are 0.3 (dotted line), 1 (dash-dot), and 3 (dashed line).

We will assume that the value of the integral is dominated by the value of V (denoted V_t) that maximizes the value of $\Phi(V, t)$ at a fixed t . We then take a second-order Taylor series approximation of $\Phi(V, t)$ around V_t :

$$\Phi(V, t) = \Phi(V_t, t) + \frac{1}{2} \Phi''(V_t, t) (V - V_t)^2 \quad (13)$$

The first-order term has vanished because V_t is a local maxima. We can then approximate equation (11) as:

$$IUH(t) \approx e^{\Phi(V_t, t)} \int_0^\infty e^{\frac{1}{2} \Phi''(V_t, t) (V - V_t)^2} dV \quad (14)$$

This integral is the well known Gaussian integral, and converges on the condition that $\Phi''(V_t, t) < 0$ to give:

$$\int_0^\infty e^{\frac{1}{2} \Phi''(V_t, t) (V - V_t)^2} dV = \frac{1}{2} \left[1 + \text{Erf} \left(V_t \sqrt{\frac{-\Phi''(V_t, t)}{2}} \right) \right] \cdot \sqrt{\frac{2\pi}{-\Phi''(V_t, t)}} \quad (15)$$

where Erf is the error function. We can find V_t by taking $0 = \partial\Phi(V, t)/\partial V$, which gives:

$$t = \frac{1}{V} + \frac{f'(V)}{f(V)} \quad (16)$$

Solving for V gives V_t . The timescale $1/V_t$ can be interpreted as the characteristic timescale of the hillslope that is dominating the recession of the impulse response at a particular point in time t .

[20] We can use this general theory to obtain expressions for the asymptotic behavior of the recession slope curves when the distribution of the 'slow' tail of hillslopes follows a power law, i.e., $f(V) \sim V^\gamma$ as $V \rightarrow 0$. The following holds true for all distributions with $\gamma > -1$. The Gamma distribution is one distribution that follows this behavior. Note that it can be shown that this is equivalent to the distribution of $T = 1/V$ going as $T^{-(\gamma+2)}$ as $T \rightarrow \infty$. The value of the characteristic timescale $1/V_t$ in such cases is given by equation (16) as:

$$1/V_t = \frac{t}{1+\gamma} \quad (17)$$

suggesting that as the catchment drains, the characteristic timescale of the hillslope dominating the discharge response increases linearly in time. The value of $e^{\Phi(V_t, t)}$ then goes as $t^{-1-\gamma}$. The value of $\Phi''(V_t, t)$ is given by:

$$\Phi''(V_t, t) = -\frac{1+\gamma}{V_t^2} \quad (18)$$

This is strictly negative only for $\gamma > -1$, which creates the restriction noted above. Substituting this into equation (15), the term in the error function becomes independent of V_t and

the integral in equation (14) goes as t^{-1} . Thus equation (14) gives the tailing behavior of the IUH as:

$$IUH(t) \sim t^{-(\gamma+2)} \quad (19)$$

That is, the tail of the instantaneous unit hydrograph follows the tail of the distribution of characteristic timescales T . If we assume that following a short period of recharge the tail of the hydrograph for a catchment follows the tail of the IUH, then $Q \sim t^{-2-\gamma}$ and $-dQ/dt \sim t^{-3-\gamma}$. Consequently it follows that:

$$-\frac{dQ}{dt} \sim Q^{1+\frac{1}{\gamma+2}} \quad (20)$$

Thus when the 'slow' tail of the distribution of hillslope characteristic timescales follows a power law with an exponent γ , the tail of the recession slope curve will also follow a power law with an exponent given by:

$$b = 1 + \frac{1}{\gamma+2} \quad (21)$$

Note that this result is in full agreement with the result for the particular case of the Gamma distribution in equation (7), where $\gamma = \alpha - 1$.

2.2.3. Bounded Power Law Distribution

[21] For distributions bounded at zero (such as the Gamma distribution), γ must be ≥ -1 for the integral of the pdf to converge. Thus the exponent b of the recession slope curve has an upper bound at 2 for such distributions. However, we can construct a distribution that permits larger b exponents for a significant part of the recession by assuming that the distribution follows a power law as V approaches zero, but is truncated at a finite value for very small V . Consider a distribution given by a power law over a bounded range of V :

$$f(V) = \frac{1}{V_0} \frac{(\gamma+1)}{1-\kappa^{\gamma+1}} \left(\frac{V}{V_0} \right)^\gamma \quad \kappa V_0 < V < V_0 \quad (22)$$

where $V_0 > 0$ is the maximum value of V and $0 < \kappa < 1$ is the size of the minimum relative to V_0 . As $\kappa \rightarrow 0$ this distribution approximates a power law pdf that goes as V^γ as $V \rightarrow 0$ (Figure 4). That is, the tails of the distribution of $T = 1/V$ go as $T^{-(\gamma+2)}$. This distribution can be substituted into equation (4) and be integrated directly to yield:

$$IUH(t) = \frac{(\gamma+1)}{\kappa^{\gamma+1}-1} (\Gamma(\gamma+2, tV_0) - \Gamma(\gamma+2, tV_0\kappa)) V_0 (tV_0)^{-\gamma-2} \quad (23)$$

where $\Gamma(x, z)$ is the incomplete Gamma function. Examples of this unit hydrograph are given in Figure 4. Again if we assume that $Q(t) \approx Q_0 \times IUH(t)$ and $Q'(t) \approx Q_0 \times IUH'(t)$ for the recession, we can plot the recession slope curve as a parametric plot of t . The expression for $IUH'(t)$ is:

$$\begin{aligned} IUH'(t) = & ((e^{-tV_0} - e^{-tV_0\kappa}) \kappa^{\gamma+2}) (tV_0)^{\gamma+2} \\ & + (\gamma+2) (\Gamma(\gamma+2, tV_0) - \Gamma(\gamma+2, tV_0\kappa)) \\ & \times \frac{(\gamma+1)}{(\kappa^{\gamma+1}-1)} V_0^2 (tV_0)^{-\gamma-3} \end{aligned} \quad (24)$$

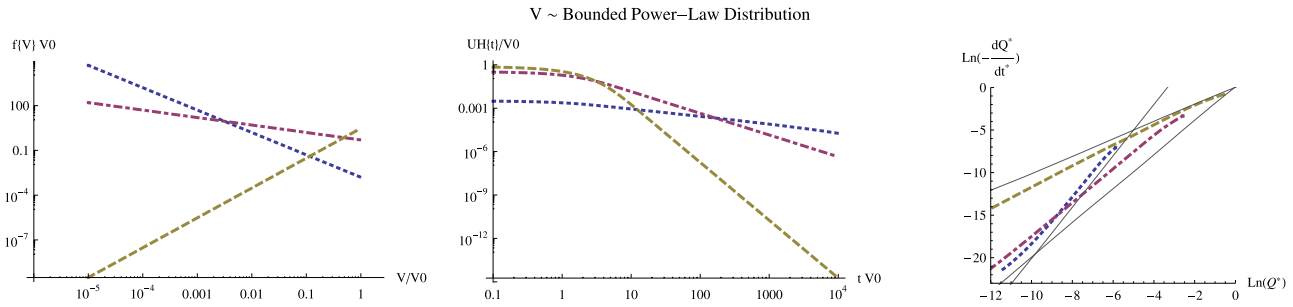


Figure 4. Same as Figure 1, but for the bounded power law distribution (equation (22)). Values of γ are -1.5 (dotted line), -0.99 (dash-dot), and 2 (dashed line). Example power laws with exponents 1, 2, and 3 are plotted in gray in Figure 4 (right).

[22] The result is shown in Figure 4. It shows clearly that exponents of b greater than 2 can occur during the recession up until a time proportional to $1/V_0$, at which point the discharge is dominated by the discharge from the slowest hillslope. The period of time over which this occurs increases with $\kappa \rightarrow 0$. In fact, for small κ the b exponent is closely approximated by equation (21), even though it was derived strictly for the case of $\gamma > -1$. Figure 5 compares the exponent predicted by that equation with the value fitted to bounded power law recession slope curves from times $t = 1/V_0$ to $t = 10/V_0$. The result suggests that equation (21) gives an upper bound to the value of b in this range of times. This bound is approached when $\kappa \rightarrow 0$.

[23] The CV of the distribution given in equation (22) can be shown to be:

$$CV = \sqrt{\frac{(\gamma+2)^2(\kappa^{\gamma+1}-1)(\kappa^{\gamma+3}-1)}{(\gamma+1)(\gamma+3)(\kappa^{\gamma+2}-1)^2}} - 1 \quad (25)$$

By combining this with equation (21) to eliminate γ , we can obtain a relationship between the CV and this ‘potential’ value of b . The relationship is shown in Figure 2 (right) for a number of different values of κ . This relationship is close to linear for small values of the CV, but for larger values it diverges and becomes dependent on the value of κ . That parameter sets the upper bound on the CV by controlling the range of values the distribution can take. The potential value of b becomes large as this upper bound is approached.

2.3. Steady State Recharge

[24] Under prolonged constant recharge, a hillslope will (in theory) approach a steady state in which the discharge is equal to the recharge. If the recharge is then stopped, the discharge will recede with a shape that is different from that of an impulse.

[25] In general we can derive the recession following the cessation of steady state recharge by convolving the impulse-response $IUH(t)$ with a step function of the form:

$$R(t) = R_0(1 - H(t)) \quad (26)$$

where $H(t)$ is the Heaviside step function and R_0 is the rate of discharge for $t < 0$. Recharge for $t > 0$ is zero. Convolution in the time domain is equivalent to multiplication in the frequency domain, so we can compute the discharge by multiplying the Fourier transfer of the impulse

response $\mathcal{F}\{IUH(t)\} = \widehat{IUH}(\omega)$ and recharge $\mathcal{F}\{R(t)\} = \widehat{R}(\omega)$ and taking the inverse transform:

$$Q = \mathcal{F}^{-1}\{\widehat{IUH} \times \widehat{R}\} \quad (27)$$

A simple closed-form solution can be found for the case of a Gamma distribution of $f(V)$. The solution is:

$$Q(t) = R_0(1 + tV_0)^{-\alpha} \quad (28)$$

Intermediate steps leading to this solution are given in Appendix A.

[26] By taking the derivative dQ/dt and eliminating the time derivative between the two equations, we can arrive at an expression for $-dQ/dt$ in terms of Q :

$$\frac{dQ}{dt} = -R_0^{-1/\alpha} V_0 \alpha Q^{1+1/\alpha} \quad (29)$$

Thus we see that the recession slope curve in this case is a power law with exponent $b = 1 + 1/\alpha$ and coefficient $a = R_0^{-1/\alpha} V_0$. Since $\alpha > 0$, the exponent b can take on any value

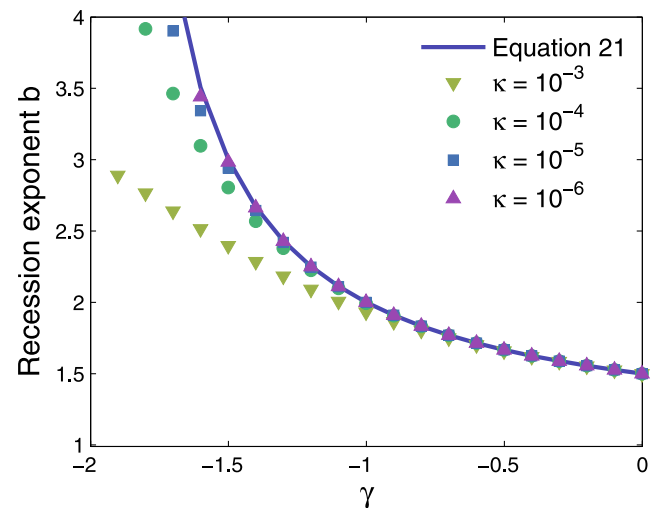


Figure 5. Comparison of the exponent of the recession slope curve predicted by equation (21) (solid line) and fitted to the early part ($t = 1/V_0$ to $10/V_0$) of the recession slope curve when $f(V)$ follows a bounded power law distribution.

greater than 1. We can invert these relationships to obtain expressions for α and V_0 in terms of a , b and R_0 :

$$\alpha = \frac{1}{b-1} \quad (30)$$

$$V_0 = a(b-1)R_0^{b-1} \quad (31)$$

Using these expressions we can calculate the mean and standard deviation of the characteristic residence times using the following expressions:

$$\mu_V = V_0\alpha = aR_0^{b-1} \quad (32)$$

$$\sigma_V = V_0\sqrt{\alpha} = aR_0^{b-1}\sqrt{b-1} \quad (33)$$

Estimation of the actual mean and variance requires that an unknown ‘effective’ steady state recharge R_0 be determined. We do not currently have a method for identifying this parameter. However, the coefficient of variation (CV) of characteristic timescales (σ_V/μ_V) in such a case is given simply by $CV = \sqrt{b-1}$, or alternatively:

$$b = 1 + CV^2 \quad (34)$$

This relationship is also plotted in Figure 2. It shows that for small variance, the difference between the exponent in the steady state case and the impulse response case is negligible for CV less than around 0.5. For larger values of CV the steady state case continues to increase, while the impulse response is bounded at $b = 2$.

2.4. Periodic Recharge

[27] The exponent of the recession slope curve for a more realistic recharge scenario (i.e., a series of storms with finite duration and interstorm period) is likely to lie somewhere between the responses for the impulse and steady state derived above. We can examine this intermediate case by assuming the recharge is periodic with a fixed duration, spacing and intensity. This simple approximation has been used by *Robinson and Sivapalan* [1997] to characterize the way the timescales of a hydrologic system (idealized as a single linear reservoir) interacted with the storm duration, intensity and interstorm period to control the flood peak.

[28] The response to periodic recharge can be found by convolving the derived impulse-response functions with the recharge time series. The example here is for the IUH with V given by a Gamma distribution. Consider a rainfall signal consisting of periods of recharge of constant intensity P_0 for duration t_r separated by an interstorm period t_b . The response of a linear system (with a known IUH function) to such a periodic forcing can be obtained from the Fourier series of the forcing. The Fourier series expansion of this rectangular wave is given by:

$$P(t) = \sum_{m=-\infty}^{\infty} D_m e^{2im\pi \frac{t}{t_r+t_b}} \quad (35)$$

where D_m is:

$$D_m = \frac{P_0}{m\pi} \sin\left(m\pi \frac{t_r}{t_r+t_b}\right) \quad (36)$$

The discharge response is obtained by multiplying each term by a transfer function $H(s)$ evaluated at $s = im\omega_p$ (where the frequency ω_p is here $2\pi/(t_r+t_b)$):

$$Q(t) = \sum_{m=-\infty}^{\infty} D_m H\left(im \frac{2\pi}{t_r+t_b}\right) e^{2im\pi \frac{t}{t_r+t_b}} \quad (37)$$

The transfer function is the Laplace transform of the impulse response function (the IUH). Using the impulse response corresponding to a Gamma distribution of V (equation (6)) this is:

$$H(s) = \alpha e^{s/V_0} \left(\frac{s}{V_0}\right)^{\alpha} \Gamma\left(-\alpha, \frac{s}{V_0}\right) \quad (38)$$

The function describing the rate of change in storage with time can be found by taking the derivative of equation (37):

$$-\frac{dQ}{dt} = \sum_{m=-\infty}^{\infty} -\frac{2\pi im}{t_r+t_b} D_m H\left(im \frac{2\pi}{t_r+t_b}\right) e^{2im\pi \frac{t}{t_r+t_b}} \quad (39)$$

We can plot the recession slope curve by evaluating equations (37) and (39) numerically (i.e., for a sufficiently large range of m around zero) for a range of values from $t = t_r/2$ to $t = t_r/2 + t_b$.

[29] Figure 6 shows the result for several values of the CV (i.e., several values of α) and t_b . These plots also show the predicted recession slope curve for the asymptotic cases of an impulse or steady state recharge with the same distribution of V . For the small CV case there is little difference between the limiting cases, and the effect of changing t_b on the exponent is negligible. For larger CV the slopes become steeper, and the effect of the interstorm period becomes pronounced. The shorter interstorm period cases behave more like the steady state recharge case and exhibit higher slopes, while the longer interstorm period cases behave more like an impulse response and take on smaller slope.

3. Application to Data

3.1. Comparison With Panola Recession Data

[30] The simple model of *Clark et al.* [2009] represents the heterogeneity in the catchment properties in terms of n discrete linear reservoirs, each with a unique timescale V_i and a fractional area F_i .

$$Q = \sum_{i=1}^n F_i V_i S_i \quad (40)$$

where S is the storage in each linear reservoir and n is the number of linear reservoirs. This model is in fact a special case of the model presented above in which $f(V)$ is a discrete distribution. That is,

$$f(V) = \sum_{i=1}^n F_i \delta(V_i) \quad (41)$$

where $\delta(V)$ is the Dirac Delta function.

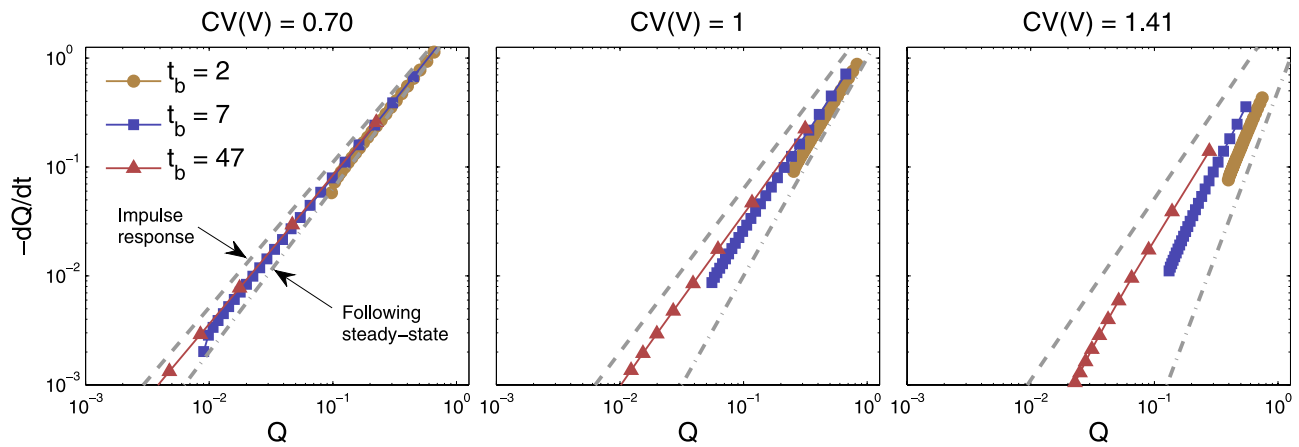


Figure 6. Sensitivity of the recession slope curve to the spacing of recharge events for Gamma distributed V . (left) When catchment variability is small, the curves are similar following an isolated impulse, following periodic recharge and following steady state recharge. (middle and right) For larger degrees of variability, the recession curves following impulse and steady state recharge become increasingly different. The recession curves following periodic recharge depend on the interstorm period, as well as the storm duration (results not shown). Results are for $\alpha = \{2, 1, 1/2\}$, $V_0 = 1 \text{ d}^{-1}$, $P_0 = 1 \text{ mm d}^{-1}$, $t_r = 3 \text{ d}$, $Q_0 = 1 \text{ mm}$, $R_0 = 1 \text{ mm d}^{-1}$.

[31] *Clark et al.* [2009] fitted this model to data from a hillslope, a 10 ha catchment and a 41 ha catchment at the Panola Mountain Research Watershed. The results are shown in Figure 7. To fit the model they assumed that the catchments had been subjected to steady state recharge of constant intensity $R_0 = 25 \text{ mm/d}$. At the hillslope scale the recession slope curve has $b \approx 1$, and they were able to fit a single linear reservoir ($n = 1$, $F_1 = 1$) with $V_1 = 2/d$. For the larger-scale catchment the recession slope curve has an exponent around $b = 2$ for large Q and $b = 1$ for lower Q . They added a second linear reservoir with $V_2 = 0.2/d$ and $F_2 = 0.2$ to approximate this behavior. At the largest scale, the recession slope data takes on a convex form, with an average slope around $b = 3$. *Clark et al.* [2009] fitted this data by reducing the fractional area of the second to $F_2 = 0.1$

and adding a third linear reservoir with $F_3 = 0.1$. They showed results for $V_3 = [0.01, 0.05, 0.1]$.

[32] We will fit the results derived above to the Panola data to demonstrate the ability of the catchment heterogeneity to explain the exponents of observed recession slope curves. Since the exponent of the recession slope curve has b as high as 3, we will only use the two cases derived above that can reproduce this behavior: (1) The bounded power law (BPL) distribution with an impulse of recharge. (2) The Gamma distribution following steady state recharge.

[33] The 10 ha and 41 ha Panola catchment have riparian aquifers and recharge from groundwater that are likely to store recharge from many previous storms in to the discharge to the stream. It is therefore unlikely that the impulse response-based BPL model will be appropriate, but it is

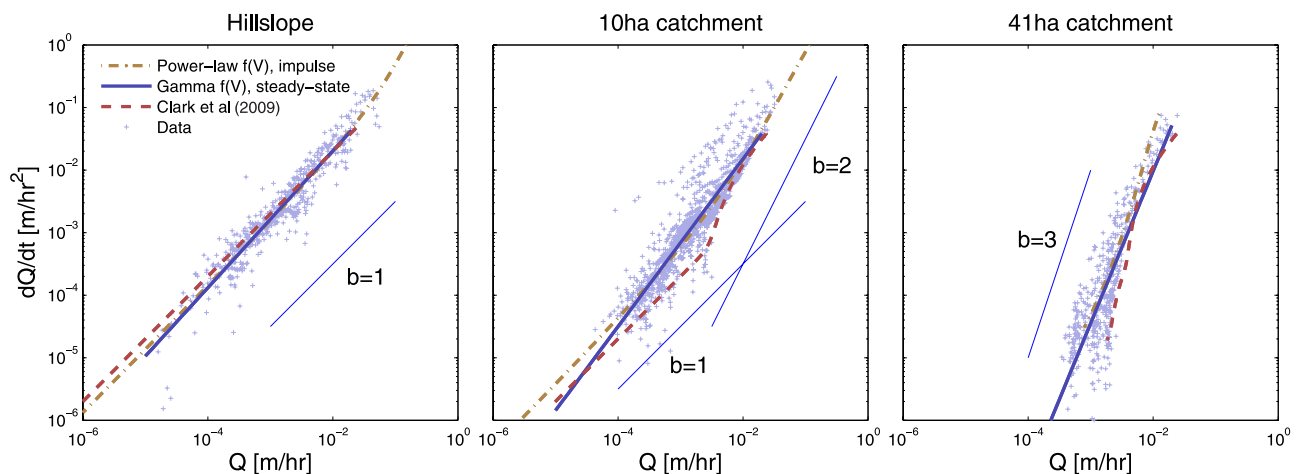


Figure 7. Recession data from the Panola Mountain Research Watershed compared to (dashed line) the multiple linear reservoir model of *Clark et al.* [2009], and the generalization to an ensemble proposed in this work with (solid line) a Gamma distribution of V and assuming initially steady state recharge, and (dash-dot line) bounded power law distribution of V and assuming an impulse of recharge with no initial storage in the catchment.

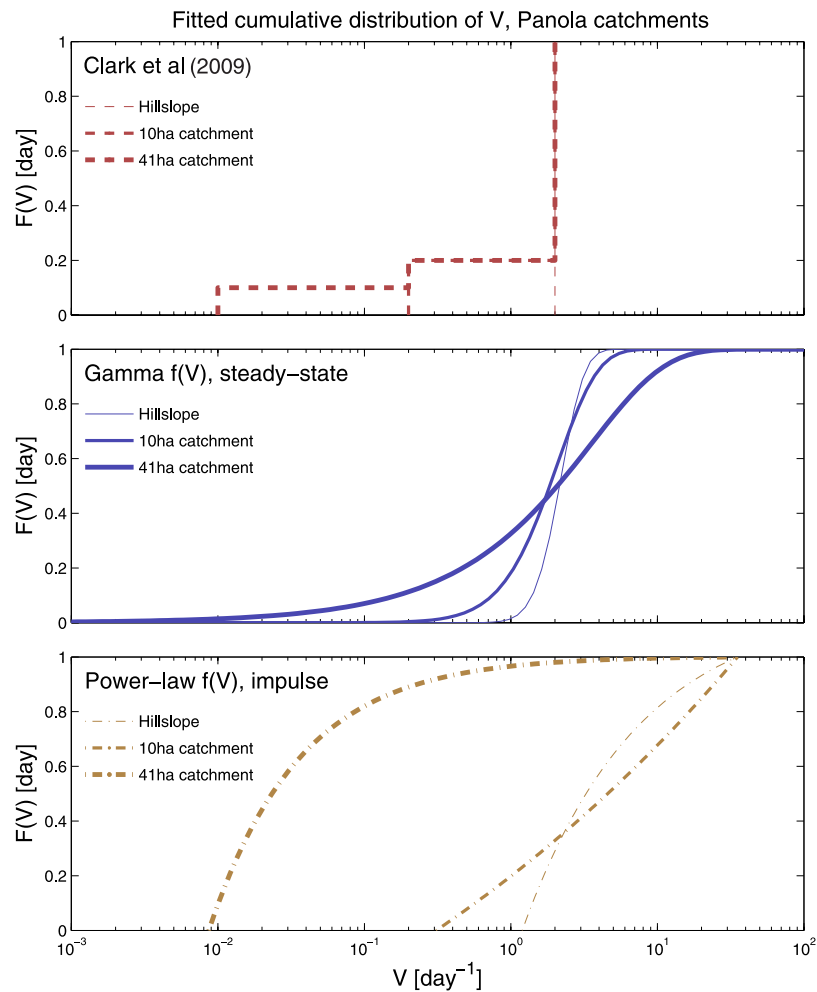


Figure 8. Comparison of the cumulative densities of V fitted to the Panola data in the work of *Clark et al.* [2009] and in this work.

included here for comparison. There is no particular reason to choose the Gamma distribution, apart from the mathematical convenience of having an analytical result to work with. However, Gamma distributions provide a launching pad for further investigation.

[34] The Gamma distribution has three parameters α , V_0 and R_0 . The value of R_0 used by *Clark et al.* [2009] was adopted (25 mm/d), and the values of α and V_0 were obtained by least squares linear regression with the log-transformed values of Q and $-dQ/dt$. The bounded power law distribution has four parameters: γ , κ , V_0 and Q_0 . A single value of Q_0 and V_0 was used across all three catchments, and γ and κ were allowed to vary. The parameters were fitted to the recession data manually.

[35] The fitted results are shown in Figure 7. In general the fit to the observed data is excellent. In all cases it is as good or better than the simple one, two or three bucket model of *Clark et al.* [2009]. The distribution parameters are given in Table 1, and the cumulative distributions for this work and *Clark et al.* [2009] are given in Figure 8.

[36] The fitted distributions generally agree with the distributions fitted by *Clark et al.* [2009] (Figure 8). The CV of the fitted Gamma distribution (equation (34)) increases from 0.31 in the hillslope to 1.20 in the 41 ha catchment, suggesting that at the hillslope scale a narrow range of characteristic timescales controls the response, and at the largest scale there is a much wider range of characteristic timescales controlling the flow. The CV of the fitted

Table 1. Parameters of the Bounded Power Law Distribution Fitted to the Panola Recession Data

	Bounded Power Law				Gamma		
	Q_0	V_0	κ	γ	R_0	V_0	α
Hillslope	68.7 mm	35.2 d ⁻¹	3.39%	-1.54	25 mm d ⁻¹	0.21 d ⁻¹	10.6
10 ha catchment	68.7 mm	35.2 d ⁻¹	0.87%	-0.88	25 mm d ⁻¹	0.71 d ⁻¹	2.9
41 ha catchment	68.7 mm	35.2 d ⁻¹	0.0246%	-1.69	25 mm d ⁻¹	5.1 d ⁻¹	0.70

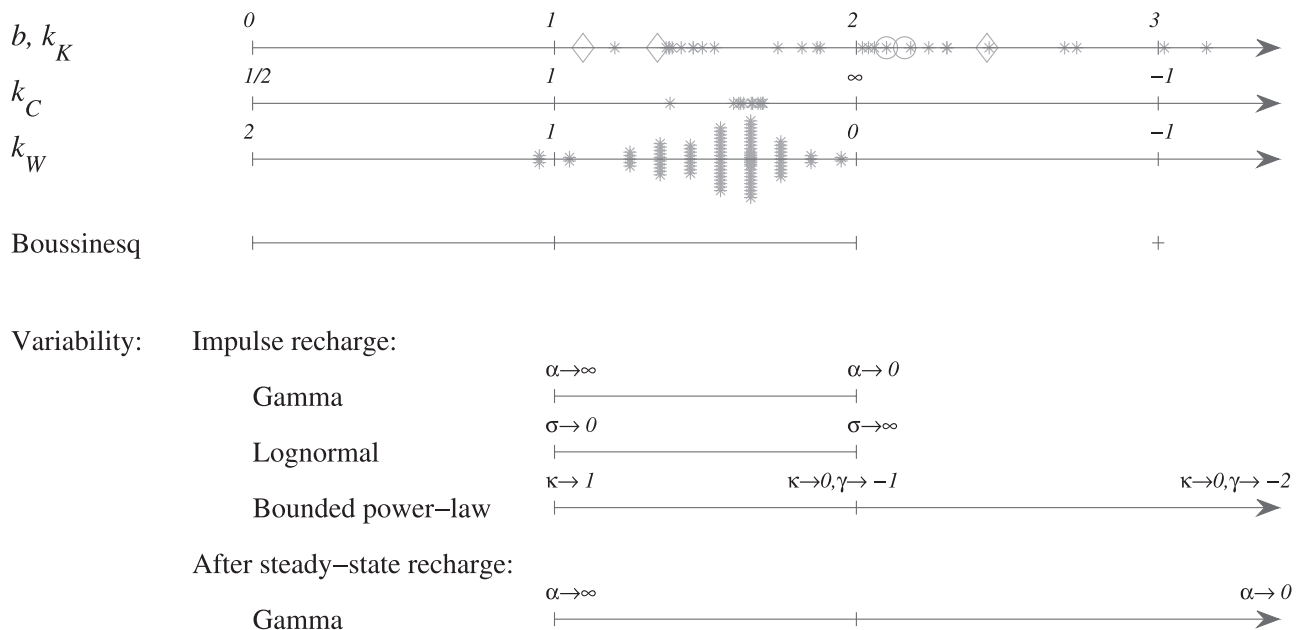


Figure 9. Comparison of observed values of the recession slope curve exponents, values obtained from approximate solutions of the Boussinesq equation, and values obtained from ensembles of linear reservoirs with various distributions of V . Values of b from Panola (diamonds) and *Tague and Grant* [2004] (stars), k_K from *Kirchner* [2009] (circles), k_C from *Chapman* [1999], k_W from *Wittenberg* [1999].

BPL distribution (equation (25)) varies from 1.1 at the small and medium scales, to 6.1 at the large scale. The spread at the large scale can be attributed to low-gradient riparian aquifers and return flow through the bedrock underlying the surrounding hillslopes. These slow-flowing areas can dominate the long-time response, while the parts of the catchment with residence times orders of magnitude shorter dominate the fast response (as suggested by *Clark et al.* [2009]).

[37] There is a striking consistency in the mean of the fitted Gamma distribution across scales. While the values of V_0 and α both vary over more than an order of magnitude, the mean V in each catchment (which is the product of V_0 and α ; see equation (32)) varies much less: Hillslope: 2.2 d^{-1} , 10 ha catchment: 2.0 d^{-1} , 41 ha catchment: 3.5 d^{-1} . It is important to note that these values are sensitive to the chosen value of R_0 , and for different values the consistency between scales is not as striking. The CV of V does not depend on R_0 (equation (32)).

3.2. Comparison to Published Data Sets of b Exponents

[38] There are numerous studies that present data on recession slope curves, and collectively they suggest that recessions from natural catchments exhibit a range of values for b . The range of values in some selected studies is shown in Figure 9. The techniques and the derivation of the reported values vary. Some studies report values of a and b fitted to the curve directly [e.g., *Tague and Grant*, 2004]. Others, including *Wittenberg* [1999] and *Chapman* [1999], have suggested that power law recession slope curves arise as a result of a nonlinear storage-discharge relationship, $Q \sim S^{1/k_W}$ or $Q \sim S^{k_C}$ respectively, and have used these models to derive expressions for the recession slope curve that could be compared to data. More recently *Kirchner* [2009] proposed analyzing $dS/dQ = g(Q)$, and found that this

function approximated a power law $g(Q) \sim Q^{k_K-1}$ (although a log quadratic function was a closer fit in at least one case). Although there are some differences in their methods, these exponents are all related to each other by:

$$b = k_K = 2 - \frac{1}{k_C} = 2 - k_W \quad (42)$$

When $b = 1$ the catchment behaves as a linear reservoir, which produces an exponential recession in time. Values of $b > 2$ imply a negative hyperbolic relationship between storage and discharge [*Kirchner*, 2009]. In addition to the observations at Panola cited here, observed values of the recession slope curve exponent b reported in the literature range from less than 1 (transformed using equation (42) from data of *Wittenberg* [1999]), to higher than 3 [*Tague and Grant*, 2004].

[39] Hydraulic theory based on the Boussinesq equation is able to account for a wide range of these exponents (Figure 9). It can be shown that a kinematic wave draining a uniformly thick sloping aquifer has $k_C = 0$ if the thickness is initially uniform and $k_C = 1/2$ (or $b = 0$) if it is initially at steady state [*Harman and Sivapalan*, 2009a; *Rupp and Selker*, 2005]. Linearized forms of the Boussinesq equation applied to the hillslope scale can produce exponents of $b = 3$ for the early stages of drainage and $b = 1$ to 2 for later stages [*Rupp and Selker*, 2006b]. *Rupp and Selker* [2005, 2006b] showed that where the hydraulic conductivity of the subsurface decays with depth as a power law, the exponent is $1 < b < 2$ and depends on the rate of decay.

[40] However, the above derivations suggest that catchment variability can also account for the same exponents. Figure 9 shows that the range of observed exponents overlaps with both the range that the heterogeneity can explain and the range that the hydraulic theory of groundwater can

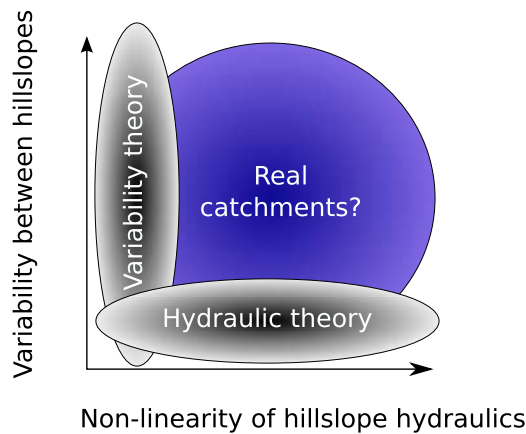


Figure 10. The hydraulic theory and the theory presented here both provide explanations for the same phenomenon, though based on different assumptions about the relative importance of the heterogeneity and hydraulics in controlling the phenomenon. Consequently, neither gives full account of the behavior of real catchments, in which both these factors are likely relevant.

explain. The ranges of exponents explained by the hydraulic theory overlaps for impulse-like recharge for Gamma and lognormal distributions for $1 < b < 2$, and the bounded power law distribution can explain observations of $b > 2$ as well. For recessions following steady state recharge the Gamma distribution can account for all $b > 1$. This overlap implies that both the hydraulics of water flow through hillslopes and the heterogeneity in catchment properties can give rise to similar signatures in the hydrograph.

4. Discussion

4.1. Recession Slope Curve as a Signature of Spatial and Temporal Variability

[41] The result presented here clearly show that the heterogeneity within catchments alone is capable of producing the apparently nonlinear, power law recession slope curves. The behavior modeled here does not arise as a result of any nonlinearity in the flow hydraulics; the underlying flow paths have been assumed to behave linearly and do not interact with each other. It is simply a result of the heterogeneity of catchment properties, represented in the distribution of timescales in the catchment. These timescales are related to the properties of the soils and the hillslope topography (equation (1)).

[42] The history of previous events appears to be an important control on the shape of the recession slope curve in heterogeneous catchments. For $f(V)$ following the Gamma distribution, the values of b for different recharge scenarios are similar for small CV , but are bounded at 2 for the impulse case, and unbounded in the steady state case (Figure 2 and equations (8) and (34)). In the more realistic situation where the recharge is intermittent over time, the observed value of b lies between the values given by these equations (Figure 6). This result suggests that we would expect the exponent b to change seasonally and interannually as the storage in the slow parts of the catchment vary.

[43] A physical interpretation of the combined control of the spatial variability of properties and the temporal variability of recharge can be constructed as follows. The discharge contributing to the recession at any particular time is dominated by different parts of the catchment as the recession progresses. As the hillslopes drain after a storm the faster hillslopes (large V) will dominate the initial response, but will drain quickly, leaving the slower (small V) hillslopes to control the later parts of the recession. This shift in dominant timescales creates the apparent nonlinearity in the recession slope curve. The rate at which control is passed from the faster to the slower part of the catchment depends on the storage in each part, which may be built up over successive recharge events. Following an impulse the storage in each part of the catchment is the same, and so the control is passed at a constant rate (equation (17)). Following an extended period of recharge events, storage is built up over time in the slower parts of the catchment (constituting a ‘memory’ of past events), which reduces the rate of change of discharge at a given discharge (Figure 6), leading to steeper recession slope curves.

4.2. Relationship to the Hydraulic Interpretation of the Recession

[44] This interpretation of the physical origins of the recession slope curve is quite different to that provided by the hydraulic theory that has often been given in the past [Brutsaert and Nieber, 1977; Brutsaert, 1994; Rupp and Selker, 2006b; Brutsaert, 1994]. In the hydraulic interpretation, the exponent of the recession slope curve arises due to the evolution of the water table gradients driving flow through a hillslope through time. If a single ‘effective’ hillslope is assumed to represent the behavior of all the hillslopes in a catchment, its properties can be determined from inversion of the empirical recession slope curves. This ‘effective’ hillslope can then be used as a basis for prediction [Zecharias and Brutsaert, 1988; Troch et al., 1993b; Pauwels et al., 2002; Szilagyi et al., 1998; Szilagyi and Parlange, 1998].

[45] The approach adopted in this work is complementary to the hydraulic approach, which ignores the variability that exists between hillslopes in a watershed and focuses attention exclusively on the flow dynamics (Figure 10). The alternative perspective examined in this work uses a simple representation of the flow dynamics in order to focus on the effect of the heterogeneity. Of course in a real catchment neither interpretation alone will do justice to reality: on the one hand the hydraulics of flow are generally more complex than a simple linear reservoir, and on the other hand catchments are highly heterogeneous. The question is more one of dominance. Under which circumstances does variability control the shape of the discharge response, and under which does the hydraulics control?

[46] However the results raise challenges for the inversion problem of determining catchment properties from hydrographs. If discharge behavior were entirely controlled by the hydraulics of flow, the recession could be used to infer hydraulic properties of catchments [e.g., Szilagyi et al., 1998]. However, because the heterogeneity of these properties can explain the same behavior, this inverse calculation cannot be achieved unless a way can be found to separate the effect of the heterogeneity. It may be that there is

insufficient information in the hydrograph alone to separate the effects of heterogeneity and hydraulics. Other information may be required, perhaps derived from spatial data or using isotope data to estimate travel time distributions.

[47] The approach used here is simplified significantly by the assumptions that the components that make up the catchment (e.g., the individual hillslopes) behave as linear systems with regards to their inputs, and are arranged in parallel. Relaxing either of these assumptions would increase the realism of the description, allowing for a more ‘physical’ description of the flow hydraulics and catchment structure, but would also make analytical results more difficult to obtain. The central points made above would remain.

5. Conclusion

[48] This work has clearly demonstrated that the exponent describing the slope of the recession slope curve, b , can be related to the variability of the hydraulic properties between hillslopes in a catchment and to the variability of recharge. By adopting a simplified representation of the flow through individual hillslopes, explicit expressions have been derived relating b to the coefficient of variation (CV) of the characteristic timescale $1/V$ for several different distributions following both impulses and steady state recharge.

[49] The derived results were found to be able to fit well to recession data from nested catchments in the Panola Mountain Research Watershed, obtained from Clark *et al.* [2009]. This allowed us to make tentative inferences about how the distribution of characteristic timescales in the landscape varied with catchment scale.

[50] It was also shown that the theory derived here can account for almost the full range of observed values of b in the literature. This suggests that both this theory and the hydraulic theory of subsurface discharge are viable explanations for the same hydrologic signature. In reality, both may be contributing to the shape of an observed recession curve.

[51] The approach presented here has enabled us to show how upscaling on the basis of the heterogeneity of landscape characteristics can manifest apparently nonlinear behavior at the catchment scale. This is an example of a catchment process being “different in form, not just different in the parameters, from the equations that describe the small-scale physics” [Kirchner, 2009, 2006; Sivapalan, 2003; Savenije, 2001]. The result gives hope that an approach can be found to capture the effects of more realistic nonlinear hydraulics and more complex spatial heterogeneity in catchment-scale models. This would further contribute toward current efforts to develop a comprehensive suite of hydrologic theory that is able to account for the effects of heterogeneity at useful scales [Blöschl and Sivapalan, 1995; McDonnell *et al.*, 2007; Troch *et al.*, 2008].

Appendix A: Key Steps Leading to Equation (28)

[52] These results were obtained using Mathematica (Wolfram Research, Inc., Mathematica, Version 6, Wolfram Research, Inc., Champaign, Illinois, 2008). To obtain the Fourier transform of the instantaneous unit hydrograph for

the case of a Gamma distribution for $f(V)$, equation (6) is first multiplied by the Heaviside step function:

$$\begin{aligned}\widehat{IUH}(\omega) &= \mathcal{F}\left\{H(t)\alpha V_0(1+tV_0)^{-1-\alpha}\right\} \\ &= \frac{\alpha|\omega|^\alpha \Gamma(-\alpha) \operatorname{sgn}(\omega) V_0^{-\alpha}}{\sqrt{2\pi}\omega} \left(|\omega| \cos\left(\frac{\pi\alpha}{2} + \frac{|\omega|}{V_0}\right) \right. \\ &\quad \left. - i\omega \sin\left(\frac{\pi\alpha}{2} + \frac{|\omega|}{V_0}\right) \right) + \frac{{}_1F_2\left(1; \frac{1}{2} - \frac{\alpha}{2}, 1 - \frac{\alpha}{2}; -\frac{\omega^2}{4V_0^2}\right)}{\sqrt{2\pi}} \\ &\quad + \frac{i\omega {}_1F_2\left(1; 1 - \frac{\alpha}{2}, \frac{3}{2} - \frac{\alpha}{2}; -\frac{\omega^2}{4V_0^2}\right)}{\sqrt{2\pi}(\alpha-1)V_0}\end{aligned}\quad (A1)$$

[53] The Fourier transform of the recharge forcing is given by:

$$\begin{aligned}\hat{R}(\omega) &= \mathcal{F}\{R_0(1-H(t))\} \\ &= R_0\left(\sqrt{\frac{\pi}{2}}\delta(\omega) - \frac{i}{\sqrt{2\pi}\omega}\right)\end{aligned}\quad (A2)$$

The result is obtained by taking the inverse transform of the product of these expressions.

[54] **Acknowledgments.** This work was supported by NSF grant ATM 06-28687. Thanks to Jake Peters (USGS), Martyn Clark and David Rupp for supplying the data from the Panola Mountain Research Watershed. The second author is grateful to Delft University of Technology for providing facilities and partial financial support during a stay as visiting professor, during which time this work was completed. Thanks also to the reviewers and the editor, Amilcare Porporato, for their detailed and helpful comments, which led to significant improvements to the manuscript.

References

- Beck, C., and E. Cohen (2003), Superstatistics, *Physica A*, 322, 267–275, doi:10.1016/S0378-4371(03)00019-0.
- Bender, C. M., and S. A. Orszag (1978), *Advanced Mathematical Methods for Scientists and Engineers*, McGraw-Hill, New York.
- Benjamin, J. R., and C. A. Cornell (1970), *Probability, Statistics, and Decision for Civil Engineers*, McGraw-Hill, New York.
- Berne, A., R. Uijlenhoet, and P. A. Troch (2005), Similarity analysis of subsurface flow response of hillslopes with complex geometry, *Water Resour. Res.*, 41, W09410, doi:10.1029/2004WR003629.
- Beven, K. (1982), On subsurface stormflow—Predictions with simple kinematic theory for saturated and unsaturated flows, *Water Resour. Res.*, 18(6), 1627–1633.
- Blöschl, G., and M. Sivapalan (1995), Scale issues in hydrological modeling: A review, *Hydrol. Processes*, 9(3–4), 251–290.
- Brutsaert, W. (1994), The unit response of groundwater outflow from a hillslope, *Water Resour. Res.*, 30(10), 2759–2763.
- Brutsaert, W., and J. Lopez (1998), Basin-scale geohydrologic drought flow features of riparian aquifers in the southern Great Plains, *Water Resour. Res.*, 34(2), 233–240.
- Brutsaert, W., and J. L. Nieber (1977), Regionalized drought flow hydrographs from a mature glaciated plateau, *Water Resour. Res.*, 13(3), 637–644.
- Chapman, T. (1999), A comparison of algorithms for stream flow recession and baseflow separation, *Hydrol. Processes*, 13(5), 701–714.
- Chapman, T. G. (2003), Modelling stream recession flows, *Environ. Modell. Software*, 18(8–9), 683–692, doi:10.1016/S1364-8152(03)00070-7.
- Clark, M. P., D. E. Rupp, R. A. Woods, H. J. T.-v. Meerveld, N. E. Peters, and J. E. Freer (2009), Consistency between hydrological models and field observations: Linking processes at the hillslope scale to hydrological responses at the watershed scale, *Hydrol. Processes*, 23(2), 311–319, doi:10.1002/hyp.7154.
- Fiori, A., and D. Russo (2007), Numerical analyses of subsurface flow in a steep hillslope under rainfall: The role of the spatial heterogeneity of the

- formation hydraulic properties, *Water Resour. Res.*, 43, W07445, doi:10.1029/2006WR005365.
- Harman, C., and M. Sivapalan (2009a), A similarity framework to assess controls on shallow subsurface flow dynamics, *Water Resour. Res.*, 45, W01417, doi:10.1029/2008WR007067.
- Harman, C., and M. Sivapalan (2009b), Effects of hydraulic conductivity variability on hillslope-scale shallow subsurface flow response and storage-discharge relations, *Water Resour. Res.*, 45, W01421, doi:10.1029/2008WR007228.
- Henderson, F. M., and R. A. Wooding (1964), Overland flow and groundwater flow from a steady rainfall of finite duration, *J. Geophys. Res.*, 69, 1531–1540.
- Kirchner, J. W. (2006), Getting the right answers for the right reasons: Linking measurements, analyses, and models to advance the science of hydrology, *Water Resour. Res.*, 42, W03S04, doi:10.1029/2005WR004362.
- Kirchner, J. (2009), Catchments as simple dynamical systems: Catchment characterization, rainfall-runoff modeling, and doing hydrology backward, *Water Resour. Res.*, 45, W02429, doi:10.1029/2008WR006912.
- Manga, M. (1999), On the timescales characterizing groundwater discharge at springs, *J. Hydrol.*, 219(1–2), 56–69, doi:10.1016/S0022-1694(99)00044-X.
- McDonnell, J. J., et al. (2007), Moving beyond heterogeneity and process complexity: A new vision for watershed hydrology, *Water Resour. Res.*, 43, W07301, doi:10.1029/2006WR005467.
- Mendoza, G., T. Steenhuis, M. Walter, and J. Parlange (2003), Estimating basin-wide hydraulic parameters of a semi-arid mountainous watershed by recession-flow analysis, *J. Hydrol.*, 279(1–4), 57–69, doi:10.1016/S0022-1694(03)00174-4.
- Moore, R. D. (1997), Storage-outflow models of streamflow recession, with application to a shallow-soil, forested catchment, *J. Hydrol.*, 198, 260–270.
- Pauwels, V. R. N., N. E. C. Verhoest, and F. P. De Troch (2002), A meta-hillslope model based on an analytical solution to a linearized Boussinesq equation for temporally variable recharge rates, *Water Resour. Res.*, 38(12), 1297, doi:10.1029/2001WR000714.
- Robinson, J. S., and M. Sivapalan (1997), Temporal scales and hydrological regimes: Implications for flood frequency scaling, *Water Resour. Res.*, 33(12), 2981–2999.
- Rupp, D., and J. Selker (2005), Drainage of a horizontal Boussinesq aquifer with a power law hydraulic conductivity profile, *Water Resour. Res.*, 41, W11422, doi:10.1029/2005WR004241.
- Rupp, D., and J. Selker (2006a), Information, artifacts, and noise in $dQ/dt-Q$ recession analysis, *Adv. Water Resour.*, 29(2), 154–160, doi:10.1016/j.advwatres.2005.03.019.
- Rupp, D. E., and J. S. Selker (2006b), On the use of the Boussinesq equation for interpreting recession hydrographs from sloping aquifers, *Water Resour. Res.*, 42, W12421, doi:10.1029/2006WR005080.
- Savenije, H. H. G. (2001), Equifinality, a blessing in disguise?, *Hydrol. Processes*, 15(14), 2835–2838.
- Shaw, S. B., A. A. Harpold, J. C. Taylor, and M. T. Walter (2008), Investigating a high resolution, stream chloride time series from the Biscuit Brook catchment, Catskills, NY, *J. Hydrol.*, 348(3–4), 245–256, doi:10.1016/j.jhydrol.2007.10.009.
- Sivapalan, M. (2003), Process complexity at hillslope scale, process simplicity at the watershed scale: Is there a connection?, *Hydrol. Processes*, 17(5), 1037–1041, doi:10.1002/hyp.5109.
- Sivapalan, M. (2005), Pattern, process and function: Elements of a unified theory of hydrology at the catchment scale, in *Encyclopedia of Hydrological Sciences*, edited by M. G. Anderson, chap. 13, John Wiley, Hoboken, N. J.
- Sloan, W. (2000), A physics-based function for modeling transient groundwater discharge at the watershed scale, *Water Resour. Res.*, 36(1), 225–241.
- Szilagyi, J., and M. Parlange (1998), Baseflow separation based on analytical solutions of the Boussinesq equation, *J. Hydrol.*, 204(1–4), 251–260.
- Szilagyi, J., M. Parlange, and J. Albertson (1998), Recession flow analysis for aquifer parameter determination, *Water Resour. Res.*, 34(7), 1851–1857.
- Tague, C., and G. Grant (2004), A geological framework for interpreting the low-flow regimes of Cascade streams, Willamette River Basin, Oregon, *Water Resour. Res.*, 40, W04303, doi:10.1029/2003WR002629.
- Tallaksen, L. M. (1995), A review of baseflow recession analysis, *J. Hydrol.*, 165(1–4), 349–370.
- Thompson, S., and G. Katul (2008), Plant propagation fronts and wind dispersal: An analytical model to upscale from seconds to decades using superstatistics, *Am. Nat.*, 171(4), 468–479, doi:10.1103/PhysRevE.71.016131.
- Touchette, H., and C. Beck (2005), Asymptotics of superstatistics, *Phys. Rev. E*, 71(1), 016131, doi:10.1086/528966.
- Troch, P., M. Mancini, C. Paniconi, and E. Wood (1993a), Evaluation of a distributed catchment scale water-balance model, *Water Resour. Res.*, 29(6), 1805–1817.
- Troch, P., F. Detroch, and W. Brutsaert (1993b), Effective water-table depth to describe initial conditions prior to storm rainfall in humid regions, *Water Resour. Res.*, 29(2), 427–434.
- Troch, P. A., G. A. Carrillo, I. Heidbüchel, S. Rajagopal, M. Switanek, T. H. M. Volkmann, and M. Yaeger (2008), Dealing with landscape heterogeneity in watershed hydrology: A review of recent progress toward new hydrological theory, *Geogr. Compass*, 2, doi:10.1111/j.1749-8198.2008.00186.x.
- Vogel, R. M., and C. N. Kroll (1992), Regional geohydrologic-geomorphic relationships for the estimation of low-flow statistics, *Water Resour. Res.*, 28(9), 2451–2458.
- Wittenberg, H. (1999), Baseflow recession and recharge as nonlinear storage processes, *Hydrol. Processes*, 13(5), 715–726.
- Zecharias, Y. B., and W. Brutsaert (1988), Recession characteristics of groundwater outflow and base-flow from mountainous watersheds, *Water Resour. Res.*, 24(10), 1651–1658.

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